

Lecture 9: Regression Models for Quantitative and Qualitative Predictors

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Outline

- 1 Polynomial regression models
- 2 Qualitative predictors
- 3 Some considerations in using indicator variables
- 4 More complex models

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Uses of polynomial models

Polynomial regression models have two basic types of uses:

- When the true curvilinear response function is indeed a polynomial function
- When the true curvilinear response function is unknown (or complex) but a polynomial function is a good approximation to the true function.

One predictor variable - second order

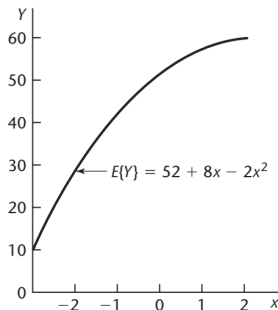
- Consider a *second-order model with one predictor variable*:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_{11} x_i^2 + \epsilon_i,$$

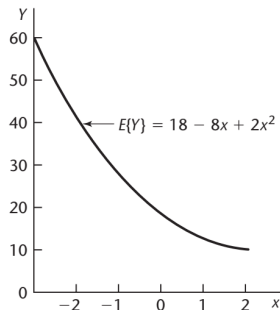
where the predictor is centered $x_i = X_i - \bar{X}$

- The response function has a quadratic form:

$$E\{Y\} = \beta_0 + \beta_1 x + \beta_{11} x^2$$



(a)



(b)

Other high order models

- One predictor variable - third order

$$Y_i = \beta_0 + \beta_1 x_i + \beta_{11} x_i^2 + \beta_{111} x_i^3 + \epsilon_i,$$

where $x_i = X_i - \bar{X}$.

- One predictor variable - higher orders
- Two predictor variables - second order

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \epsilon_i,$$

where $x_{i1} = X_{i1} - \bar{X}_1$ and $x_{i2} = X_{i2} - \bar{X}_2$.

- Fitting of polynomial regression models are special cases of the general linear regression model.

Case example - life of a type power cell

- A researcher studied the effects of the charge rate and temperature on the life of a new type of power cell in a preliminary small-scale experiment. The charge rate (X_1) was controlled at three levels (.6, 1.0, and 1.4 amperes) and the ambient temperature (X_2) was controlled at three levels (10, 20, 30°C). The life of the power cell (Y) was measured in terms of the number of discharge-charge cycles that a power cell underwent before it failed.
- The researcher was not sure about the nature of the response function in the range of the factors studied, hence he/she decided to fit the second-order polynomial regression model:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \epsilon_i,$$

Case example - life of a type power cell

Cell	(1) Number of Cycles	(2) Charge Rate	(3) Temperature	(4)	(5)	(6)	(7)	(8)
i	Y_i	X_{i1}	X_{i2}	<u>Coded Values</u>				
				x_{i1}	x_{i2}	x_{i1}^2	x_{i2}^2	$x_{i1} x_{i2}$
1	150	.6	10	-1	-1	1	1	1
2	86	1.0	10	0	-1	0	1	0
3	49	1.4	10	1	-1	1	1	-1
4	288	.6	20	-1	0	1	0	0
5	157	1.0	20	0	0	0	0	0
6	131	1.0	20	0	0	0	0	0
7	184	1.0	20	0	0	0	0	0
8	109	1.4	20	1	0	1	0	0
9	279	.6	30	-1	1	1	1	-1
10	235	1.0	30	0	1	0	1	0
11	224	1.4	30	1	1	1	1	1
		$\bar{X}_1 = 1.0$	$\bar{X}_2 = 20$					

Case example - life of a type power cell

- X_1 and X_2 are centered and scaled:

$$x_{i1} = \frac{X_{i1} - \bar{X}_1}{0.4} = \frac{X_{i1} - 1.0}{0.4}; \quad x_{i2} = \frac{X_{i2} - \bar{X}_2}{10} = \frac{X_{i2} - 20}{10}$$

(the denominator used for each predictor variable is the absolute difference between adjacent levels of the variable)

- Correlations:

$$\sigma(X_1, X_1^2) = 0.991; \quad \sigma(x_1, x_1^2) = 0.0$$

$$\sigma(X_2, X_2^2) = 0.986; \quad \sigma(x_2, x_2^2) = 0.0$$

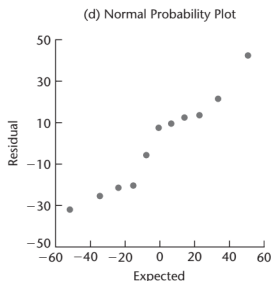
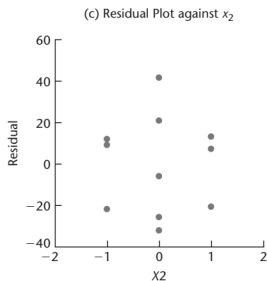
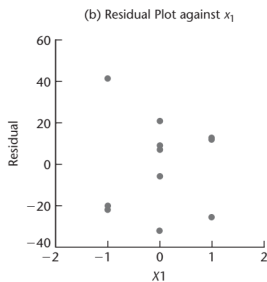
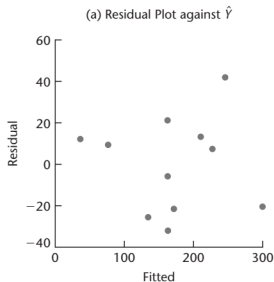
Low levels of multicollinearity can be helpful in avoiding computational inaccuracies.

Case example - life of a type power cell

- Fitting of Model

$$\hat{Y} = 162.84 - 55.83x_1 + 75.50x_2 + 27.39x_1^2 - 10.61x_2^2 + 11.50x_1x_2.$$

Case example - Residual plots



None of these plots suggest any gross inadequacies of regression model

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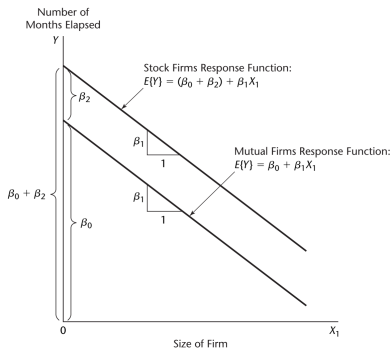
Qualitative predictors

- Many predictor variables of interest in business, economics, and the social and biological sciences are qualitative.
- Examples of qualitative predictor variables are gender (male, female), purchase status (purchase, no purchase), and disability status (not disabled, partly disabled, fully disabled)

Qualitative predictor with two classes

- A good presentation: A qualitative variable with c classes will be represented by $c - 1$ indicator variables, each taking on the values 0 and 1.
- In a study of innovation in the insurance industry, an economist wished to relate the speed with which a particular insurance innovation is adopted (Y) to the size of the insurance firm (quantitative) and the type of firm (stock or mutual company, qualitative).
- Denote variables X_{i1} : size of firm i and $X_{i2} = 1$ if stock company and $X_{i2} = 0$ otherwise.
- The response function for this regression model is:
$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Qualitative predictor with two classes - interpretation



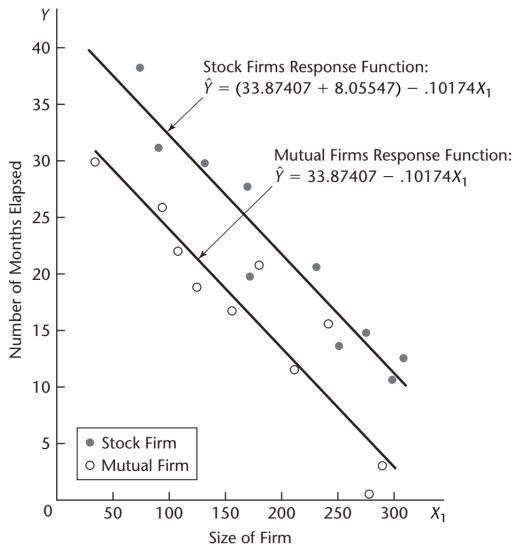
- The mean time elapsed before the innovation is adopted, $E\{Y\}$, is a linear function of size of firm (X_1), with the same slope β_1 for both types of firms.
- β_2 shows how much higher (lower) the mean response line is for the class coded 1 than the line for the class coded 0, for any given level of X_1 .

Qualitative predictor with two classes - Example

	(1)	(2)	(3)	(4)	(5)
Firm	Number of	Size of Firm	Type of	Indicator	
i	Months Elapsed	(million dollars)	Firm	Code	$X_{i1} X_{i2}$
	Y_i	X_{i1}		X_{i2}	
1	17	151	Mutual	0	0
2	26	92	Mutual	0	0
3	21	175	Mutual	0	0
4	30	31	Mutual	0	0
5	22	104	Mutual	0	0
6	0	277	Mutual	0	0
7	12	210	Mutual	0	0
8	19	120	Mutual	0	0
9	4	290	Mutual	0	0
10	16	238	Mutual	0	0
11	28	164	Stock	1	164
12	15	272	Stock	1	272
13	11	295	Stock	1	295
14	38	68	Stock	1	68
15	31	85	Stock	1	85
16	21	224	Stock	1	224
17	20	166	Stock	1	166
18	13	305	Stock	1	305
19	30	124	Stock	1	124
20	14	246	Stock	1	246

Qualitative predictor with two classes - Example

Fitted response function is: $\hat{Y} = 33.87407 - 0.10174X_1 + 8.05547X_2$



Qualitative predictor with two classes - Example

- The economist was most interested in the effect of type of firm (X_2) on the elapsed time and wished to obtain a 95 percent confidence interval for β_2 .
- We have $t(0.975; 17) = 2.110$ and the confidence limits $8.05547 \pm 2.110 * 1.45911$. Therefore, the confidence interval for β_2 is: $4.98 \leq \beta_2 \leq 11.13$.
- With 95 percent confidence, we conclude that stock companies tend to adopt the innovation somewhere between 5 and 11 months later, on the average, than mutual companies, for any given size of firm.

Qualitative predictor with more than two classes

- Consider the regression of tool wear (Y) on tool speed (X_1) and tool model, where the latter is a qualitative variable with four classes (M_1, M_2, M_3, M_4).
- Tool model:
 - $X_2 = 1$ if tool model M1, and 0 otherwise
 - $X_3 = 1$ if tool model M2, and 0 otherwise
 - $X_4 = 1$ if tool model M3, and 0 otherwise
- The data input for the X variables

Tool model	X_1	X_2	X_3	X_4
M1	X_{i1}	1	0	0
M2	X_{i1}	0	1	0
M3	X_{i1}	0	0	1
M4	X_{i1}	0	0	0

- First-order model is:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \epsilon_i$$

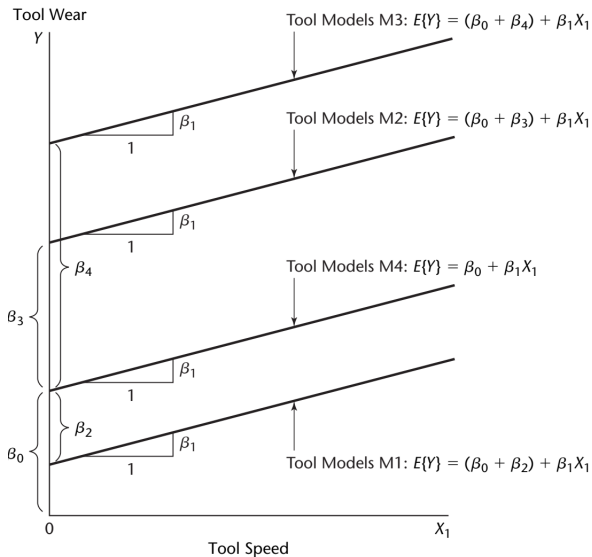
Tool wear example

General response function:

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Response function for tool model M4: $E\{Y\} = \beta_0 + \beta_1 X_1$
- Response function for tool model M1: $E\{Y\} = (\beta_0 + \beta_2) + \beta_1 X_1$
- Response function for tool model M2: $E\{Y\} = (\beta_0 + \beta_3) + \beta_1 X_1$
- Response function for tool model M3: $E\{Y\} = (\beta_0 + \beta_4) + \beta_1 X_1$

Tool wear example



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Indicator variables versus allocated codes

Allocated codes:

- The predictor variable “frequency of product use” which has three classes: frequent user, occasional user, nonuser.
- Using a single X_1 variable is employed and values are assigned to the classes: X_1 is 3 for frequent user, 2 for occasional user, and 1 for non user.
- The allocated codes are, of course, arbitrary and could be other sets of numbers.
- The first-order model with allocated codes is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \epsilon_i$$

- From above model:

$$\begin{aligned}E\{Y|\text{frequent user}\} - E\{Y|\text{occasional user}\} = \\E\{Y|\text{occasional user}\} - E\{Y|\text{nonuser}\} = \beta_1\end{aligned}$$

- The coding 1, 2, 3 implies that the mean response changes by the same amount when going from a nonuser to an occasional user as when going from an occasional user to a frequent user.

Indicator variables versus allocated codes

- Indicator variables, in contrast, make no assumptions about the spacing of the classes and rely on the data to show the differential effects that occur.
- For example, two indicator variables, using two variables X_1 and X_2 , are employed to represent the qualitative variable, as follows: (X_1, X_2) for frequent user, occasional user and nonuser are $(1, 0)$, $(0, 1)$ and $(0, 0)$, respectively.
- The first-order regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i$$

- β_1 measures the differential effect:
 $E\{Y|\text{frequent user}\} - E\{Y|\text{nonuser}\}$
- β_2 measures the differential effect:
 $E\{Y|\text{occasional user}\} - E\{Y|\text{nonuser}\}.$
- $\beta_1 - \beta_2$ measures the differential effect between frequent user and occasional user.

Indicator variables versus quantitative variables

- Indicator variables can be used even if the predictor variable is quantitative. For instance, the quantitative variable age may be transformed by grouping ages into classes such as under 21, 21–34, 35–49, *etc.*

Other codings for indicator variables

- Alternative coding 1: X_2 is 1 if stock company and -1 if mutual company.
 - The response function: $E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
 - The response function for stock firm: $E\{Y\} = (\beta_0 + \beta_2) + \beta_1 X_1$
 - The response function for mutual firm: $E\{Y\} = (\beta_0 - \beta_2) + \beta_1 X_1$
- Alternative coding 2 (one-hot-encoding): Use a c indicator variables for c classes of the qualitative variable.
 - Model is $Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \epsilon_i$, where X_{i1} is size of firm; $X_{i2} = 1$ if stock company and 0 otherwise; $X_{i3} = 1$ if mutual company and 0 otherwise.
 - Two response functions: $E\{Y\} = \beta_0 + \beta_2 + \beta_1 X_1$ for stock firm and $E\{Y\} = \beta_0 + \beta_3 + \beta_1 X_1$ for mutual firm.
(We can drop the intercept term in the linear regression model.)

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More than one qualitative predictor variable

- Consider the regression of advertising expenditures (Y) on sales (X_1), type of firm (incorporated, not incorporated), and quality of sales management (high, low).
- Define $X_2 = 1$ if firm incorporated and 0 otherwise; $X_3 = 1$ if quality of sales management high and 0 otherwise.
- First-order model:

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \epsilon_i$$

- First-order model with interactions:

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i1} X_{i2} + \beta_5 X_{i1} X_{i3} + \beta_6 X_{i2} X_{i3} + \epsilon_i$$

THANK YOU!

Chapter 8, Kutner textbook.

Chapters 7-8, MPV, Introduction to linear regression analysis